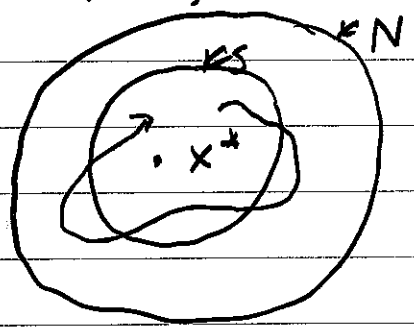
Side notes relevant to Problem 2 on PS#1

$$\frac{dX}{dt} = AX \qquad \text{vs} \qquad \frac{dX}{dt} = A(t)X$$

constant matrix — T-periodic matrix

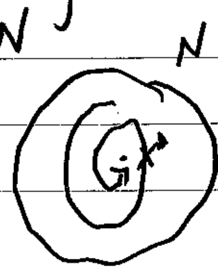
You are asked about "stability properties" of $X=0$. Here are a few definitions for this

An equilibrium x^* (e.g. $x^*=0$) is Lyapunov stable if for every neighborhood N of x^*, there exists a neighborhood $S \subset N$ s.t. for all $x(0) \in S$, $x(t) \in N$ for all $t \geq 0$.



unstable: not Lyapunov stable (there are some N for which S doesn't exist)

asymptotically stable: Lyapunov stable & there exists a neighborhood N s.t. $\lim_{t\to\infty} x(t) = x^*$ for all $x(t) \in N$



If $\dot{X} = AX$, A = constant matrix with eigenvalues λ with $Re(\lambda) < 0$, then $X = 0$ is asymptotically stable equilibrium soln. If $Re(\lambda) > 0$ for any eigenvalue, then $X = 0$ is unstable.

Contrast with

$$\frac{dX}{dt} = A(t)X, \quad \text{with} \quad A(t) = A(t+T)$$

[Note: in PS#1, I call $A(t)$ by $M(t)$, which may confuse people who think that the monodromy matrix should be M]

eigenvalues of $A(t)$ do not determine stability of $X=0$. There is another matrix, called the "Monodromy matrix" (denoted M) that has eigenvalues μ ("Floquet multipliers") that determine stability of $X=0$.

Consider fundamental matrix soln $\Phi(t)$ that satisfies

$$\frac{d\Phi}{dt} = A(t)\Phi, \quad \Phi(0) = Id.$$

Then $\Phi(T) = M$ = monodromy matrix

initially $X(0) = X_0$, then at time T, $X(T) = MX_0$, $X(2T) = M^2X_0$, etc.

If all eigenvalues μ_j of M satisfy $|\mu_j| < 1$, then $X = 0$ is stable equilibrium

If any eigenvalue μ_j of M has $|\mu| > 1$, then $X = 0$ is unstable

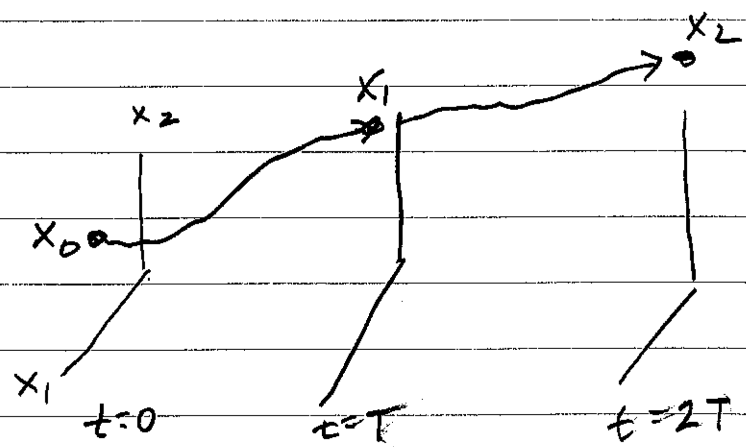
$$X(nT) = M^n X_0$$

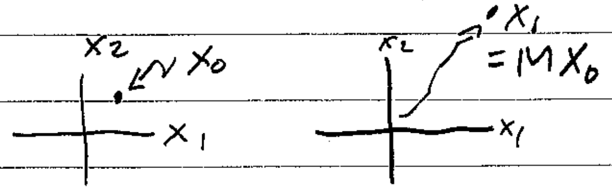
$$X_0 = \sum_j c_j v_j \qquad \text{where } M v_j = \mu_j v_j$$

$$\Rightarrow X(nT) = \sum_j \mu_j^n c_j v_j$$

if $|\mu_j| > 1$ then $|\mu_j|^n \to \infty$ as $n \to \infty$.

if $|\mu_j| < 1$ then $|\mu_j|^n \to 0$ as $n \to \infty$





etc.

end of side note

Chapter 3: existence & uniqueness of solns of ODE's

$$(*)\begin{cases} \dot{x} = V(x) \\ X(0) = X_0 \end{cases} \qquad \begin{aligned} &V : \mathbb{R}^n \to \mathbb{R}^n \\ &X_0 \in \mathbb{R}^n \end{aligned}$$

existence thm

If V is continuous then a soln exists on some interval of time containing $t=0$.

Need not exist for all time. Finite time blow-up is possible, where $|X(t)| \to \infty$ as $t \to T$, the blow-up time.

Note: we do not require V to be differentiable for existence.

$$X(t) = X_0 + \int_0^t V(x(s))\,ds$$

solves $(*)$, easy to check.

uniqueness theorem

Suppose there are $a, b > 0$ s.t. $V : B_b(X_0) \to \mathbb{R}^n$ is Lipschitz with constant K. Then $(*)$ has a unique soln $X(t)$ for $t \in J = [-a, a]$ provided $a = b/M$ where $M = \max_{x \in B_b(X_0)} |V(x)|$

$B_b(X_0)$ = ball of radius b about X_0
$= \{X : |X - X_0| \le b\}$

$t \in J = [-a, a]$, $a = b/M$, ensures that soln. doesn't escape the ball in the time interval J.

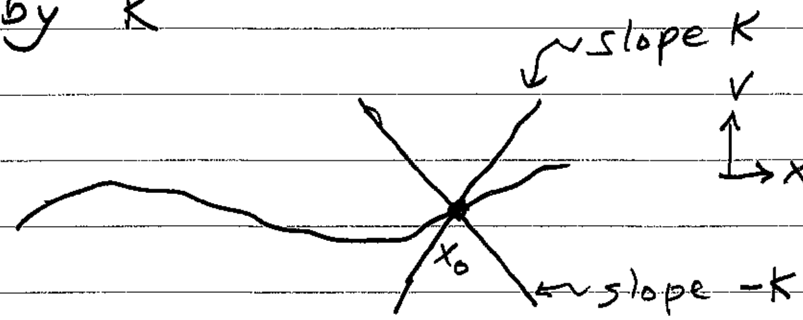
<u>Lipschitz with constant K</u>

$$|V(X_2) - V(X_1)| \le K\,|X_2 - X_1| \qquad \forall X_1, X_2 \in B_b(X_0)$$

i.e.
$$\frac{|V(X_2) - V(X_1)|}{|X_2 - X_1|} \le K \qquad (X_1 \ne X_2)$$

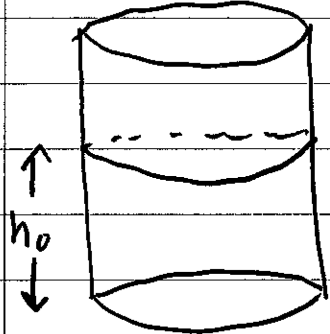
absolute value of slope of secant line is bounded by K

$X \in \mathbb{R}$ case

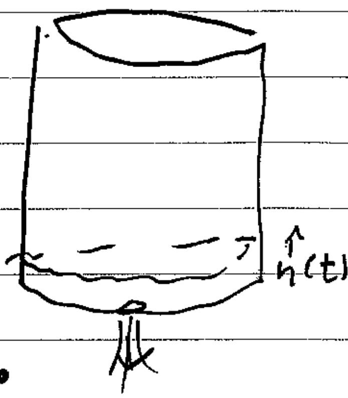


If $V(X)$ is C^1, i.e. differentiable, then it is Lipschitz. If it is Lipschitz then it's C^0, but it need not be differentiable, e.g.
$\dot{x} = |x|$ is Lipschitz continuous with $K = 1$

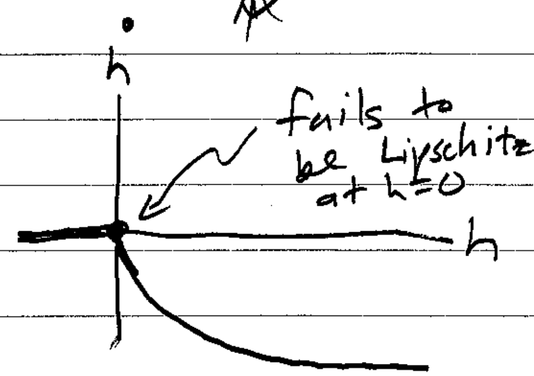
example 1 : shows that lack of uniqueness can be "physical" - not necessarily a flaw of the model



at $t=0$, punch hole in bucket



$$\frac{dh}{dt} = \begin{cases} -\sqrt{h} & , \ h > 0 \\ 0 & , \ h = 0 \end{cases}$$



Construct soln for $h_0 > 0$

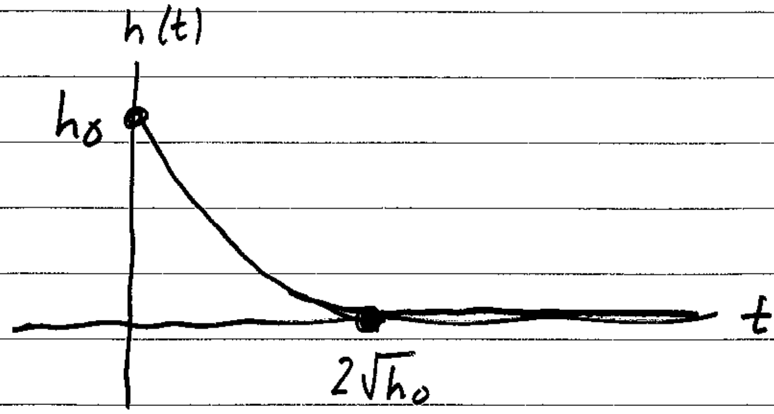
$$\frac{dh}{dt}\left(\frac{1}{-\sqrt{h}}\right) = 1$$

$$\int_0^t \frac{dh}{ds}\left(-\frac{1}{\sqrt{h}}\right) ds = \int_0^t ds$$

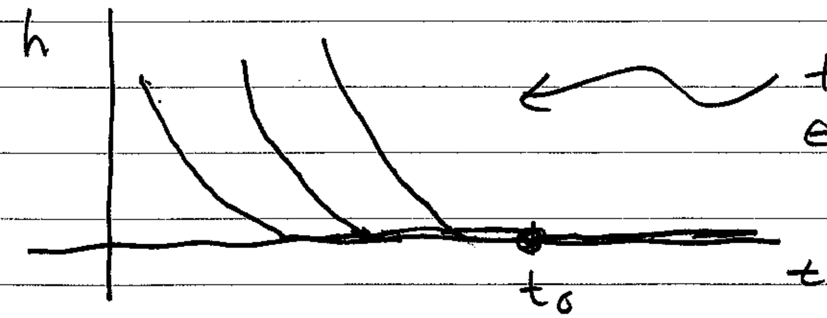
$$\int_{h_0}^{h(t)} -\frac{1}{\sqrt{h}} \, dh = t \Rightarrow \underbrace{-2\sqrt{h(t)}}_{h(t) \geq 0} + 2\sqrt{h_0} = t$$

$$t \in [0, t_{max}], \ t_{max} = 2\sqrt{h_0}$$

$$h(t) = \begin{cases} \left(\sqrt{h_0} - \frac{t}{2}\right)^2 & t < 2\sqrt{h_0} \\ 0 & t \geq 2\sqrt{h_0} \end{cases}$$



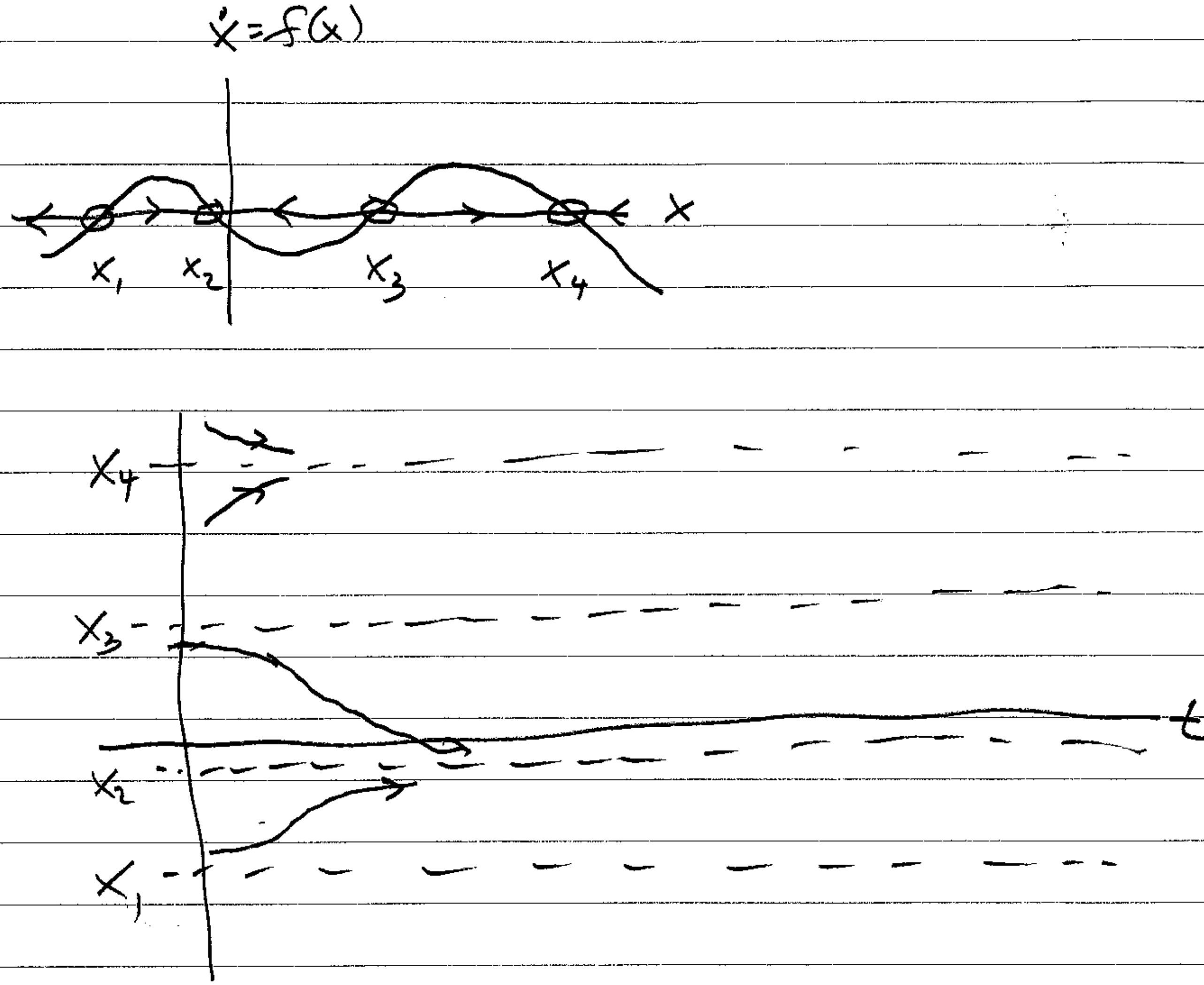
Note soln. is not unique on a neighborhood of $t = t_0$ for $t_0 > 2\sqrt{h_0}$



Family of solns, each with a different $h_0 = h(0) \geq 0$ up to some largest h_0

example 2

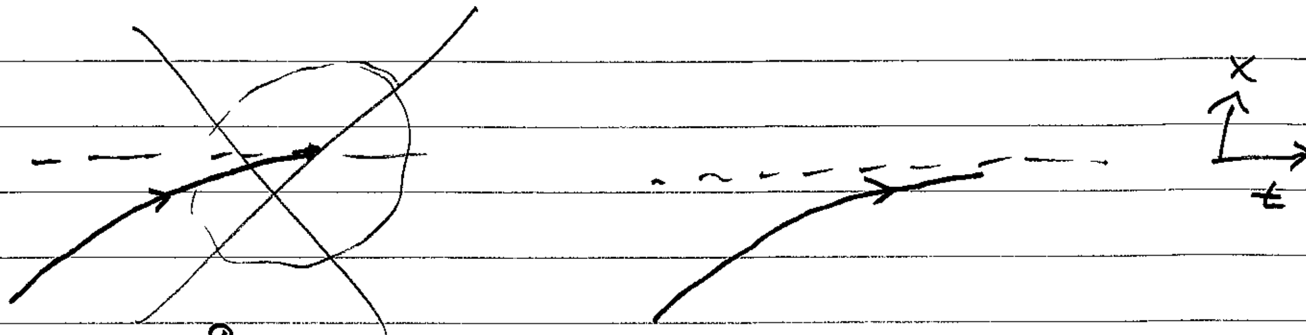
Qualitative analysis of $\dot{x} = f(x)$, $x \in \mathbb{R}$



f is smooth $\Rightarrow$ solns are unique

For $x_0 \in [x_1, x_4]$ $x(t) \to x^*$, where $x^* \in \{x_1, x_2, x_3, x_4\}$

How long does it take to get to x^* if it doesn't start at x^*? Must take infinite amount of time. ~~Unlike~~ bucket problem



violates
uniqueness of solns.